

RDP+ v1.0.9

A lightweight Windows connection launcher for RDP, HTTP/HTTPS, and SSH targets — built in C++17 with the native Win32 API for zero-dependency deployment.

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Compiler: MSVC 19.51 (Visual Studio 2022)

Features

- Single-pane list view of all saved connections
 - Supports RDP, HTTP, HTTPS, and SSH protocols
 - Double-click or press Enter to launch any connection
 - Click column headers to sort by Host, Type, or Description
 - "New RDP Connection" button opens `mstsc.exe` directly
 - SSH connections prompt for a username and launch in Windows Terminal (falls back to `cmd.exe`)
 - JSON-based connection list — easy to edit, no database required
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Requirements

- Windows 10 or later
 - Windows SDK (Common Controls v6)
 - `json.hpp` — nlohmann/json single-header (included alongside source)
 - `resource.h` + `RDP_Plus.res` — application icon
-

Build

```
CL /EHsc /W3 /O2 /GL /DUNICODE /D_UNICODE /DNDEBUG /std:c++17 RDP_Plus.cpp
RDP_Plus.res ^
/Fe:RDP_Plus.exe /link /SUBSYSTEM:WINDOWS ^
comctl32.lib shell32.lib shlwapi.lib user32.lib gdi32.lib
```

Flag	Purpose
<code>/O2 /GL</code>	Speed optimisation + whole-program optimisation
<code>/DUNICODE /D_UNICODE</code>	Full Unicode (UTF-16) throughout
<code>/SUBSYSTEM:WINDOWS</code>	No console window
<code>/std:c++17</code>	Required for structured bindings in JSON parsing

connections.json

Place `connections.json` in the **same directory** as `RDP_Plus.exe`. Two layouts are accepted:

Array format (recommended)

```
json
[
  { "host": "server1.local", "type": "RDP", "description": "Domain controller" },
  { "host": "192.168.1.10", "type": "SSH", "description": "Linux build server" },
  { "host": "intranet.corp", "type": "HTTPS", "description": "Internal web portal" }
]
```

Keyed-object format

```
json
{
  "dc": { "host": "server2.local", "type": "RDP", "description": "File server" },
  "cameras": [
    { "host": "10.0.0.5", "type": "HTTP", "description": "Camera NVR" }
  ]
}
```

Supported `type` values

type	Behaviour
<code>RDP</code>	Launches <code>mstsc.exe /v:<host></code>
<code>HTTP</code>	Opens <code>http://<host></code> in default browser

type	Behaviour
HTTPS	Opens <code>https://<host></code> in default browser
SSH	Prompts for username, runs <code>ssh -p 22 <user>@<host></code>

Type matching is case-insensitive.

Usage

1. Edit `connections.json` with your hosts.
2. Run `RDP_Plus.exe`.
3. Double-click a row (or select it and press **Enter**) to launch.
4. Click **New RDP Connection** to open the standard Windows Remote Desktop dialog.
5. Click **Exit** (or close the window) to quit.

Column headers are clickable — click once to sort ascending, again to reverse.

Project Structure

```
RDP_Plus.cpp      # All application source
RDP_Plus.res      # Compiled resource file (icon)
resource.h        # Resource ID definitions
json.hpp          # nlohmann/json single-header library
connections.json  # Your connection list (not included – create your own)
```

Notes

- **SSH port** is hardcoded to 22. Non-standard ports are not currently supported via `connections.json`.
- **Windows Terminal** (`wt.exe`) is tried first for SSH sessions; `cmd.exe` is the fallback.
- `WM_LAUNCH` (`WM_USER + 1`) is defined but reserved for a future async launch feature.
- The `UsernameDialog` struct in the source is superseded by the inline dialog proc in `GetUsernamePopup` and can safely be removed.

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